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Question 1

Not answered

Marked out of 1.00

• Ref: Bateman et al. BMC Psychiatry. 2016; 16(1): 304. • To investigate whether Mentalization-based treatment (MBT) can be effective and acceptable for alleviating symptoms of antisocial personality disorder (ASPD), a sub-sample of patients were recruited from a randomised trial that originally intended to treat suicidality, self-harm, and borderline personality disorder (BPD). Of the 52 patients enrolled, 12 refused randomisation, leaving 40 with a dual diagnosis of ASPD and BPD, who were randomised to one of the two outpatient treatment programmes. The study investigated whether outpatients with comorbid borderline personality disorder and ASPD receiving MBT (n=21) were more likely to show improvements in symptoms related to aggression than those offered a structured protocol (n= 19) of similar intensity but excluding MBT components. All patients were offered approximately 140 sessions of therapy in total. Seventy-five percent across the two groups met criteria for completers - at least 70 sessions attended over the first year. There was no difference in the distribution of completer categories across the groups ($\chi^2 = 1.87$, $df = 2$). Phase 1 of the study involved collecting qualitative data using a semistructured interview with respect to the patients' experience of undergoing the treatments. Results strongly suggested that the patients valued the intervention, and were particularly appreciative of the group format. The phase 2 (quantitative analysis) found benefits from MBT for ASPD-associated behaviours in patients with comorbid BPD and ASPD, including the reduction of anger, hostility, paranoia, and frequency of self-harm and suicide attempts, as well as the improvement of depression measured using Beck's Depression Inventory (group difference at 18 months [95% CI] = -8.96 [-14.27, -3.65]). Which of the following methods has been employed in this study?

Select one:

- ☐ Meta-analysis
- ☐ Qualitative method
- ☐ Mixed methods
- ☐ Quantitative method
- ☐ Economic analysis

Your answer is incorrect.

The term "mixed methods" is used to refer to a methodical amalgamation of both quantitative and qualitative data within a single research inquiry. This approach permits a more thorough utilization of data in a synergistic manner than do separate quantitative and qualitative studies. Excerpt from <https://pcmh.ahrq.gov>

The correct answer is: Mixed methods



Question 2

Not answered

Marked out of 1.00

• Ref: Bateman et al. BMC Psychiatry. 2016; 16(1): 304. • To investigate whether Mentalization-based treatment (MBT) can be effective and acceptable for alleviating symptoms of antisocial personality disorder (ASPD), a sub-sample of patients were recruited from a randomised trial that originally intended to treat suicidality, self-harm, and borderline personality disorder (BPD). Of the 52 patients enrolled, 12 refused randomisation, leaving 40 with a dual diagnosis of ASPD and BPD, who were randomised to one of the two outpatient treatment programmes. The study investigated whether outpatients with comorbid borderline personality disorder and ASPD receiving MBT (n=21) were more likely to show improvements in symptoms related to aggression than those offered a structured protocol (n= 19) of similar intensity but excluding MBT components. All patients were offered approximately 140 sessions of therapy in total. Seventy-five percent across the two groups met criteria for completers - at least 70 sessions attended over the first year. There was no difference in the distribution of completer categories across the groups ($\chi^2 = 1.87$, $df = 2$). Phase 1 of the study involved collecting qualitative data using a semistructured interview with respect to the patients' experience of undergoing the treatments. Results strongly suggested that the patients valued the intervention, and were particularly appreciative of the group format. The phase 2 (quantitative analysis) found benefits from MBT for ASPD-associated behaviours in patients with comorbid BPD and ASPD, including the reduction of anger, hostility, paranoia, and frequency of self-harm and suicide attempts, as well as the improvement of depression measured using Beck's Depression Inventory (group difference at 18 months [95% CI] = -8.96 [-14.27, -3.65]). What can you conclude from this study?

Select one:

- ☐ MBT is a feasible option to treat depression in the presence of ASPD
- ☐ MBT is an effective treatment for BPD
- ☐ MBT is an effective treatment for ASPD
- ☐ MBT is a feasible option to treat BPD in the presence of ASPD
- ☐ MBT is a feasible option to treat ASPD in the presence of BPD

Your answer is incorrect.

The main aim was to investigate whether Mentalization-based treatment (MBT) can be effective and acceptable for alleviating symptoms of antisocial personality disorder and involved subjects with a diagnosis of borderline personality disorder. The results suggest that both qualitatively and quantitatively, MBT is a feasible option to treat ASPD in the presence of BPD.

The correct answer is: MBT is a feasible option to treat ASPD in the presence of BPD



Question 3

Not answered

Marked out of 1.00

• Ref: Bateman et al. BMC Psychiatry. 2016; 16(1): 304. • To investigate whether Mentalization-based treatment (MBT) can be effective and acceptable for alleviating symptoms of antisocial personality disorder (ASPD), a sub-sample of patients were recruited from a randomised trial that originally intended to treat suicidality, self-harm, and borderline personality disorder (BPD). Of the 52 patients enrolled, 12 refused randomisation, leaving 40 with a dual diagnosis of ASPD and BPD, who were randomised to one of the two outpatient treatment programmes. The study investigated whether outpatients with comorbid borderline personality disorder and ASPD receiving MBT (n=21) were more likely to show improvements in symptoms related to aggression than those offered a structured protocol (n= 19) of similar intensity but excluding MBT components. All patients were offered approximately 140 sessions of therapy in total. Seventy-five percent across the two groups met criteria for completers - at least 70 sessions attended over the first year. There was no difference in the distribution of completer categories across the groups ($\chi^2 = 1.87$, $df = 2$). Phase 1 of the study involved collecting qualitative data using a semistructured interview with respect to the patients' experience of undergoing the treatments. Results strongly suggested that the patients valued the intervention, and were particularly appreciative of the group format. The phase 2 (quantitative analysis) found benefits from MBT for ASPD-associated behaviours in patients with comorbid BPD and ASPD, including the reduction of anger, hostility, paranoia, and frequency of self-harm and suicide attempts, as well as the improvement of depression measured using Beck's Depression Inventory (group difference at 18 months [95% CI] = -8.96 [-14.27, -3.65]). Which of the following is an important part of processing the data collected using a semi-structured interview?

Select one:

- ☐ Using a standard set of follow-up questions to ensure reliability
- ☐ Keeping hand-written notes of the interview to enable content analysis
- ☐ Scoring the responses in a Likert fashion
- ☐ Recording of the interview for a detailed analysis
- ☐ Weighting the scores for follow-up questions differently from the core questions

Your answer is incorrect.

The recording of a semistructured interview makes it easier for the researcher to focus on the interview content and the verbal prompts and thus enables the transcriptionist to generate "verbatim transcript" of the interview. Excerpt from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4194943/>

The correct answer is: Recording of the interview for a detailed analysis



Question 4

Not answered

Marked out of 1.00

• Ref: Bateman et al. BMC Psychiatry. 2016; 16(1): 304. • To investigate whether Mentalization-based treatment (MBT) can be effective and acceptable for alleviating symptoms of antisocial personality disorder (ASPD), a sub-sample of patients were recruited from a randomised trial that originally intended to treat suicidality, self-harm, and borderline personality disorder (BPD). Of the 52 patients enrolled, 12 refused randomisation, leaving 40 with a dual diagnosis of ASPD and BPD, who were randomised to one of the two outpatient treatment programmes. The study investigated whether outpatients with comorbid borderline personality disorder and ASPD receiving MBT (n=21) were more likely to show improvements in symptoms related to aggression than those offered a structured protocol (n= 19) of similar intensity but excluding MBT components. All patients were offered approximately 140 sessions of therapy in total. Seventy-five percent across the two groups met criteria for completers - at least 70 sessions attended over the first year. There was no difference in the distribution of completer categories across the groups ($\chi^2 = 1.87$, $df = 2$). Phase 1 of the study involved collecting qualitative data using a semistructured interview with respect to the patients' experience of undergoing the treatments. Results strongly suggested that the patients valued the intervention, and were particularly appreciative of the group format. The phase 2 (quantitative analysis) found benefits from MBT for ASPD-associated behaviours in patients with comorbid BPD and ASPD, including the reduction of anger, hostility, paranoia, and frequency of self-harm and suicide attempts, as well as the improvement of depression measured using Beck's Depression Inventory (group difference at 18 months [95% CI] = -8.96 [-14.27, -3.65]). Which of the following is a particular strength of this study?

Select one:

- ☐ Both qualitative and quantitative data have been gathered
- ☐ The follow-up is long enough to demonstrate that the differential effects are maintained
- ☐ The study is well-powered to demonstrate group differences
- ☐ The sample is very representative of the wider anti-social personality disorder population
- ☐ The study has used a double-blind treatment approach

Your answer is incorrect.

This study is not sufficiently powered to demonstrate a conclusive result; it is based on a sub-sample from another study. The results support that MBT is acceptable and broadly feasible to be offered in the treatment of ASPD features seen in patients with BPD; but the follow-up is too short to establish durability of these benefits. While there is no information about blinding, it is unlikely that the patients

were fully blind to the given therapy. One positive aspect of the study is the acquisition of both quantitative and qualitative data.
The correct answer is: Both qualitative and quantitative data have been gathered



Question 5

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Ref: Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table.

With respect to epilepsy, if the CIDT results are positive, this increases the probability of being epileptic by

Select one:

- ☐ 7.2 times
- ☐ 98%
- ☐ 77%
- ☐ 44%
- ☐ 9.88 times

Your answer is incorrect.

The positive likelihood ratio (LR+) refers to the increase in the probability of disease if the test is positive (mathematically, this is the true positivity rate divided by false positivity rate). For epilepsy this is 9.88

The correct answer is: 9.88 times



Question 6

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Ref: Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table.

Table 2
Psychometric properties

	<i>n</i>	%	<u>PPV</u>	<u>NPV</u>	LR+	LR-
All respondents	195	100	0.64	0.93	2.71	0.12
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

- PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Overall, how many of the expected positives have a confirmed disorder?

Select one:

- ☐ 12%
☐ 64%
☐ 2.17%
☐ 93%
☐ 0.60%

Your answer is incorrect.

This number is given by the overall PPV of 0.64 (see the first row of the table). Approximately two-thirds of the CIDT positives have a confirmed disorder.

The correct answer is: 64%



Question 7

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Ref: Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table.

With respect to alcohol use disorder, how many of those expected to be unaffected on the basis of CIDT are in fact free of the diseases listed here?

Select one:

- ☐ 93%
- ☐ 98%
- ☐ 64%
- ☐ 12%
- ☐ 2.71%

Your answer is incorrect.

This number is given by the NPV of 0.98 (see the last row of the table). Approximately 98% of the CIDT epilepsy-negatives are not epileptic.

The correct answer is: 98%



Question 8

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Ref: Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table. For which condition listed in this table are the positively screened subjects most likely to have the condition tested?

Select one:

- ☐ Epilepsy
- ☐ Depression
- ☐ Conduct disorder
- ☐ Alcohol use disorder
- ☐ Psychoses

Your answer is incorrect.

If the screening test is positive, the probability that a test-positive subject actually has the disease is given by PPV. This is highest for depression.

The correct answer is: Depression



Question 9

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table. In which group of patients does a negative test notably reduce the probability of being diagnosed with the condition when a diagnostic interview is performed by clinicians?

Select one:

- ☐ Psychoses
- ☐ Conduct disorder
- ☐ Epilepsy
- ☐ Alcohol use disorder
- ☐ Depression

Your answer is incorrect.

This question relates to the value of a negative test result in changing the pre-test probability of the diagnosis. In other words, this refers to negative likelihood ratio (LR-). With a very small negative likelihood ratio, the post-test probability of epilepsy in those with a negative CIDD result is likely to be small.

The correct answer is: Epilepsy



Question 10

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Ref: Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table. The real-life use of the CIDT focuses on reporting only the positive test results by the untrained informants. In this context, the key indicator of diagnostic utility of the CIDT is

Select one:

- ☐ The odds ratio
- ☐ The PPV
- ☐ The NPV
- ☐ The LR-ve
- ☐ The accuracy

Your answer is incorrect.

The reported real-life use of the CIDI only results in reporting positives; in other universal screening procedures, both negative and positive results are considered to be equally meaningful. In this case, the key indicator of accuracy of the CIDI are the PPV and the positive likelihood ratio.

The correct answer is: The PPV



Question 11

Not answered

Marked out of 1.00

Table 2 Psychometric properties

	<i>n</i>	%	PPV ^a	NPV ^a	LR+	LR-
Disorder-specific analysis ^b						
Depression	45	23.1	0.53	0.87	3.87	0.49
Epilepsy	14	7.2	0.44	0.98	9.88	0.23
Psychoses (adult only)	20	10.3	0.32	0.92	3.67	0.64
Conduct disorder (child only)	11	5.6	0.28	0.89	2.00	0.61
Alcohol use disorder	11	5.6	0.33	0.98	8.11	0.30

PPV, positive predictive value; NPV, negative predictive value; LR+, positive likelihood ratio; LR-, negative likelihood ratio.

Ref: Jordans et al., The British Journal of Psychiatry (2015) 207, 501-506. To evaluate the accuracy of a community-based proactive case-finding strategy (Community Informant Detection Tool, CIDT), involving pictorial vignettes in primary care settings, untrained community informants were employed to use the CIDT and identify screen positive ($n = 110$) and negative persons ($n = 85$). Participants were then administered the Composite International Diagnostic Interview (CIDI) by clinicians. Overall, the CIDT has a positive predictive value of 0.64 (0.68 for adults only) and a negative predictive value of 0.93 (0.91 for adults only). The results for individual disorders are shown in the attached table. Which of the following statements regarding diagnostic test performance is true?

Select one:

- ☐ NPV decreases if there is a decrease in the prevalence of a condition
- ☐ NPV and PPV are not affected by changing the target population in a screening program
- ☐ NPV increases if the screening program targets individuals at a higher risk of a condition
- ☐ PPV increases if the screening program targets individuals at a higher risk of a condition
- ☐ PPV increases if there is a decrease in the prevalence of a condition

Your answer is incorrect.

Predictive values are determined by the sensitivity and specificity of the test and the prevalence of disease in the population being tested. The more sensitive a test, the less likely an individual with a negative test will have the disease and thus the greater the negative predictive value. The more specific the test, the less likely an individual with a positive test will be free from disease and the greater the positive predictive value. Excerpt from: <https://www.health.ny.gov/diseases/chronic/discreen.htm>

The correct answer is: PPV increases if the screening program targets individuals at a higher risk of a condition



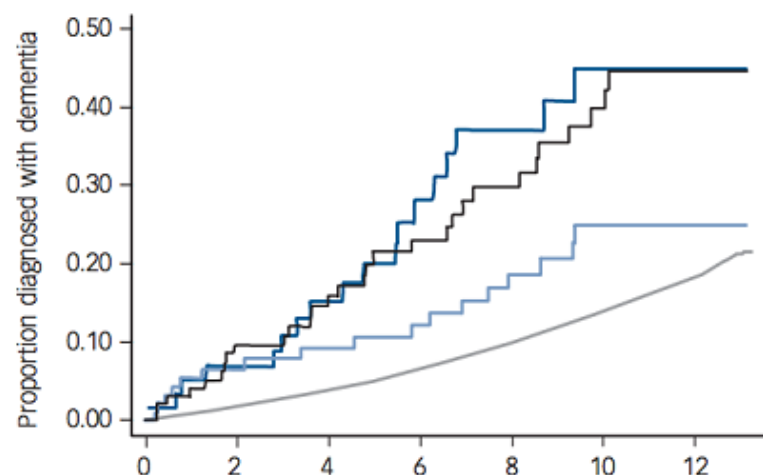
Question **12**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 1.51, 95% CI 1.28-1.77). The attached graph is likely to

Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.26
Diseases of the brain	3758 (9.9)	34 (12.1)	1.12	0.72-1.69
Bipolar disorder	302 (0.8)	8 (3.1)	3.66	1.81-7.40
Respiratory diseases	1636 (4.3)	10 (3.9)	0.86	0.46-1.60
Liver and digestive diseases	568 (1.5)	8 (3.1)	2.02	1.00-4.06
Kidney diseases	421 (1.1)	4 (1.6)	1.35	0.50-3.63
Other causes	Men's S520 (1.4)	5 (1.9)	1.36	0.56-3.29

Select one:

- ☐ Frequency histogram

- ☐ ROC curve

a. Mortality sub-hazard ratio derived from an age-adjusted competing risk regression model.

Results in bold are statistically significant.

- ☐ KP graph

- ☐ Stem-leaf plot

- ☐ Kaplan Meier Curve

Your answer is incorrect.

The figure depicts age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder. When the hazard (probability of an event) is plotted against time, instead of the survival (probability of not experiencing the event), the Kaplan Meier curve ascends up instead of descending down.

The correct answer is: Kaplan Meier Curve

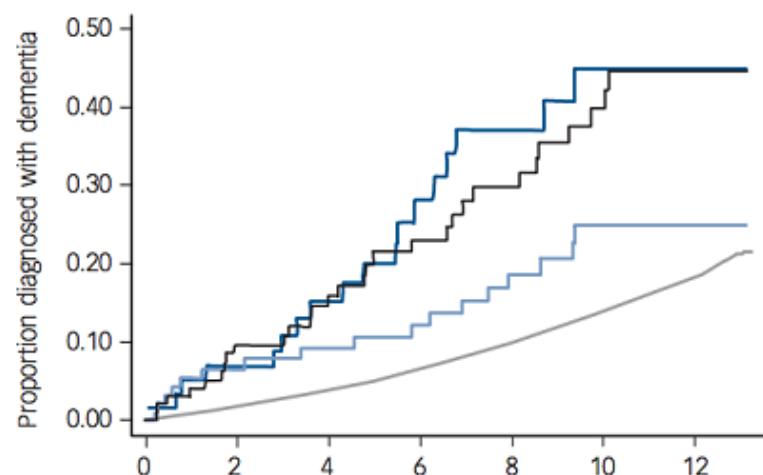
Question **13**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 1.51, 95% CI 1.28-1.77). Which of the following is the most common cause of death among those with bipolar disorder in this study?

Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.28
Diseases of the brain	3758 (10.0)	38 (17.1)	0.12	0.72-1.99
Infectious diseases	3021 (8.0)	11 (5.0)	1.1	0.25-5.62
Pneumonia or influenza	1190 (3.1)	8 (3.1)	3.66	1.81-7.40
Respiratory diseases	1636 (4.3)	10 (3.9)	0.86	0.46-1.60
Liver and digestive diseases	568 (1.5)	8 (3.1)	2.02	1.00-4.06
Kidney diseases	421 (1.1)	4 (1.6)	1.35	0.50-3.63
Other causes	Men's S520 (1.4)	5 (1.9)	1.36	0.56-3.29

Select one:

- ☐ Cancer
- ☐ Brain diseases
- ☐ Accident

☐ Infection

☐ Suicide

a. Mortality sub-hazard ratio derived from an age-adjusted competing risk regression model.

Results in bold are statistically significant.

Your answer is incorrect.

The table shows that bipolar disorder is associated with increased hazard of death by suicide, accidents, pneumonia or influenza, and to some extent, diseases of the liver and digestive system. Nevertheless, cancer is the most common cause of death, killing 16% of bipolar patients.

The correct answer is: Cancer

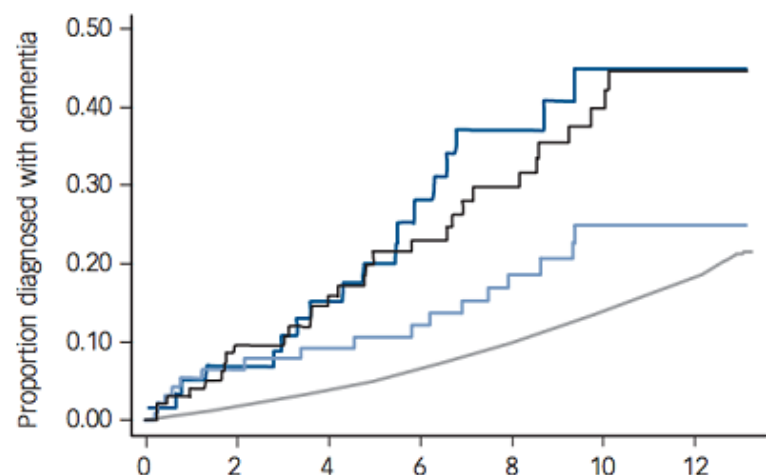
Question **14**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 3.66, 95% CI 3.12-4.32). Which of the following causes of death is most significantly increased among those with bipolar disorder in this study?

Cancer	4484 (11.9)	41 (10.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.28
Diseases of the brain	375 (1.0)	34 (12.1)	0.12	0.07-0.19
Infection	302 (0.8)	8 (3.1)	1.61	0.72-3.62
Suicide	1636 (4.3)	10 (3.9)	3.66	1.81-7.40
Respiratory diseases	568 (1.5)	8 (3.1)	0.86	0.46-1.60
Liver and digestive diseases	421 (1.1)	4 (1.6)	2.02	1.00-4.06
Kidney diseases	Men's S520 (1.4)	5 (1.9)	1.35	0.50-3.63
Other causes			1.36	0.56-3.29

Select one:

- ☐ Suicide
- ☐ Infection
- ☐ Accident

a. Mortality sub-hazard ratio derived from an age-adjusted competing risk regression model.

Results in bold are statistically significant.

- ☐ Brain diseases
- ☐ Cancer

Your answer is incorrect.

The table shows that bipolar disorder is associated with increased hazard of death by suicide, accidents, pneumonia or influenza, and to some extent, diseases of the liver and digestive system. Nevertheless, suicide has the highest hazard ratio (13.43 times more likely in bipolar disorder).

The correct answer is: Suicide

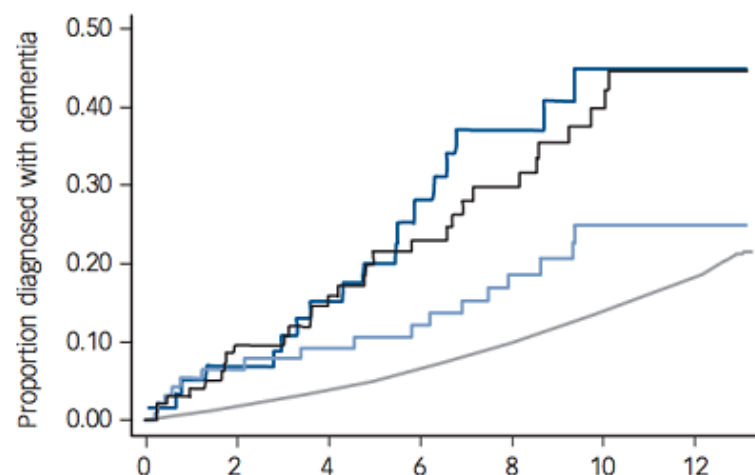
Question **15**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 2.31, 95% CI 1.28-1.77). Which of the following is accurate interpretation from the attached graph?

	Men's 55-64	Men's 65-74	Men's 75-84	Men's 85+
Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.26
Diseases of the brain	3958 (10.5)	38 (12.1)	1.12	0.72-1.69
Bipolar disorder	302 (1.4)	11 (2.3)	1.51	1.28-1.77
Other causes	319 (0.8)	8 (3.1)	3.66	1.81-7.40
Respiratory diseases	1636 (4.3)	10 (3.9)	0.86	0.46-1.60
Liver and digestive diseases	568 (1.5)	8 (3.1)	2.02	1.00-4.06
Kidney diseases	421 (1.1)	4 (1.6)	1.35	0.50-3.63
Men's 55-64	5520 (1.4)	5 (1.9)	1.36	0.56-3.29

Select one:

- ☐ The risk of dementia is greatest among those with no history of bipolar disorder
- ☐ Loss to follow-up is greatest among those with <5 years of bipolar disorder
- ☐ There is a significant increase in the age-adjusted risk of dementia among those with 5-14 years of bipolar disorder
- ☐ The risk of dementia is greatest among those with <5 years of bipolar disorder
- ☐ The risk of dementia is greatest among those with >15 years of bipolar disorder

Your answer is incorrect.

Compared to those with no bipolar disorder, 2 groups of patients [<5 years of illness (HR of 2.69), >15 years (HR of 2.68)] have an increased age-adjusted risk of dementia. Patients with 5-14 years of illness have a HR of 1.41, but this is not significant (CI 0.86 to 2.31).

The correct answer is: The risk of dementia is greatest among those with <5 years of bipolar disorder

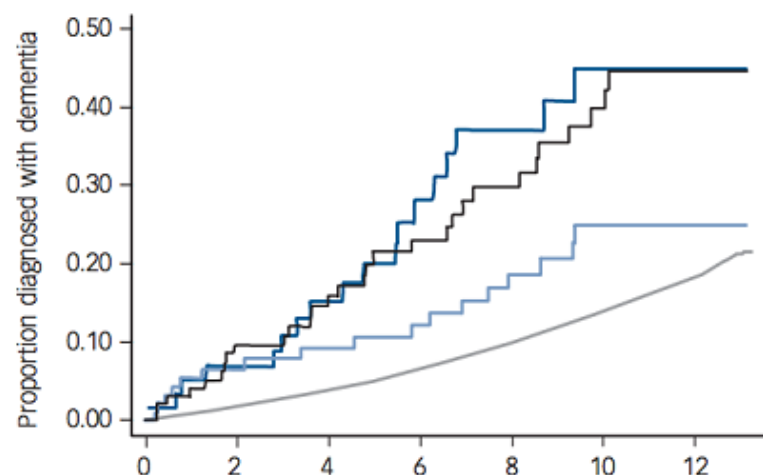
Question **16**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 3.15, 95% CI 1.28-1.77). What can you conclude from the results presented in this study? Bipolar disorder

Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.28
Diseases of the brain	3958 (10.5)	34 (12.1)	1.12	0.72-1.69
Bipolar disorder	3021 (8.0)	8 (3.1)	3.66	1.81-7.40
Presented in this study	319 (0.8)	10 (3.9)	0.86	0.46-1.60
Respiratory diseases	1636 (4.3)	8 (3.1)	2.02	1.00-4.06
Liver and digestive diseases	568 (1.5)	4 (1.6)	1.35	0.50-3.63
Kidney diseases	421 (1.1)	5 (1.9)	1.36	0.56-3.29
Other causes	Men's 5520 (1.4)			

Select one:

- ☐ Increases the risk of dementia and premature death
- ☐ Increases the risk of dementia but reduces the risk of premature death
- ☐ Results in bold are statistically significant
- ☐ Reduces the risk of dementia but increases the probability of premature death
- ☐ Leads to premature ageing of an individual
- ☐ Should be rigorously treated to reduce the risk of dementia

Your answer is incorrect.

The presented results show that older adults with a past diagnosis of bipolar disorder are at increased risk of dementia and that this risk is greatest for those with long-standing and with recent history of bipolar disorder. In a study assessing the utility of alcohol screening tools, 242 medical inpatients received the Alcohol Use Disorders Identification Test (AUDIT), CAGE' and brief Michigan Alcoholism Screening Test (brief MAST) questionnaires. Sensitivities when identifying ICD-10 based harmful use pattern were 93, 79 and 35%, respectively. Specificity values were 94, 86 and 97%, respectively. Positive predictive values were 74, 44 and 69% respectively.

The correct answer is: Increases the risk of dementia and premature death

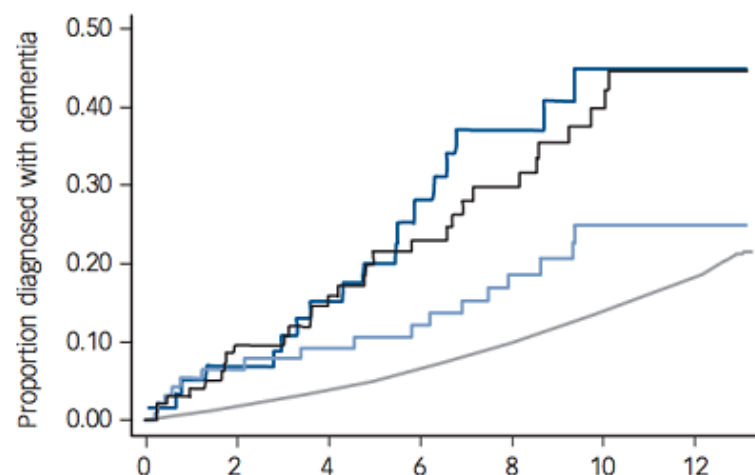
Question **17**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 2.31, 95% CI 1.28-1.77). Based on the results above, the prevalence of MDD in the UK is given by

Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.26
Diseases of the brain	3758 (10.0)	34 (12.1)	1.12	0.72-1.69
Respiratory diseases	1636 (4.3)	10 (3.9)	0.86	0.46-1.60
Liver and digestive diseases	568 (1.5)	8 (3.1)	2.02	1.00-4.06
Kidney diseases	421 (1.1)	4 (1.6)	1.35	0.50-3.63
Other causes	Men's S520 (1.4)	5 (1.9)	1.36	0.56-3.29

Select one:

☐ 4%

☐ 7%

☐ 21%

a. Mortality sub-hazard ratio derived from an age-adjusted competing risk regression model.

Results in bold are statistically significant.

☐ 20%

☐ Cannot be determined

Your answer is incorrect.

This cannot be determined from the data provided to us.

The correct answer is: Cannot be determined

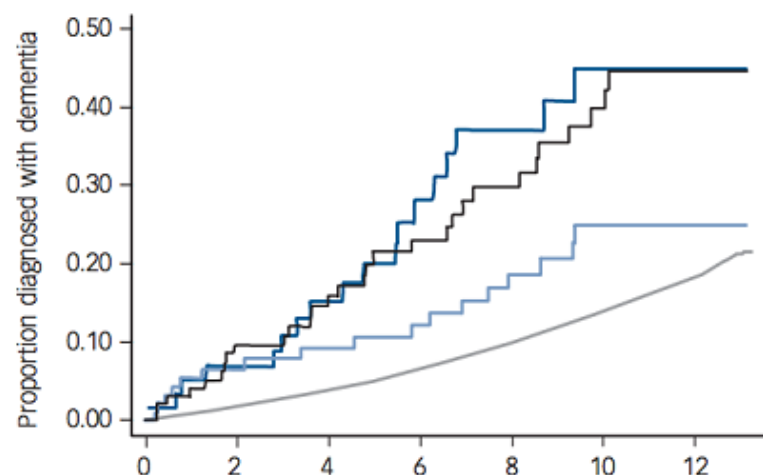
Question **18**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

Ref: Almeida et al., The British Journal of Psychiatry (2016) 209, 121-126. To investigate the 13-year risk of dementia and death in older adults with bipolar disorder, a cohort study of 37 768 men aged 65-85 years was conducted. The outcomes dementia and death were traced using electronic record linkage. Bipolar disorder was associated with increased adjusted hazard ratio (HR) of dementia (HR 2.30, 95% CI 1.80-2.94). Bipolar disorder was also associated with increased mortality (HR 2.31, 95% CI 1.28-1.77).

What is the most important aspect of the study design that can ensure the validity of the reported results?

Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.26
Diseases of the brain	3954 (10.5)	34 (12.1)	1.12	0.72-1.69
Bipolar disorder	502 (1.4)	8 (3.1)	3.66	1.81-7.40
Respiratory diseases	1636 (4.3)	10 (3.9)	0.86	0.46-1.60
Liver and digestive diseases	568 (1.5)	8 (3.1)	2.02	1.00-4.06
Kidney diseases	421 (1.1)	4 (1.6)	1.35	0.50-3.63
Other causes	Men's S520 (1.4)	5 (1.9)	1.36	0.56-3.29

- Select one:
- ☐ The tests were fully computerised
 - ☐ There was no drop-outs in the study
 - ☐ a. Mortality sub-hazard ratio derived from an age-adjusted competing risk regression model.
 - ☐ Results in bold are statistically significant
 - ☐ Subjects were randomly allocated to the three tests

- ☐ The reference standard was applied independent of the individual test results
- ☐ The patients were unaware of their individual scores

Your answer is incorrect.

The performance of a diagnostic test is determined by comparison with a gold standard. An ideal reference standard should be applied to all test subjects irrespective of whether they test positive or negative to the diagnostic tests being evaluated. Furthermore, the persons applying and interpreting the diagnostic test must be kept blind to the results of the comparison standard. Excerpt from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3263223/>

The correct answer is: The reference standard was applied independent of the individual test results

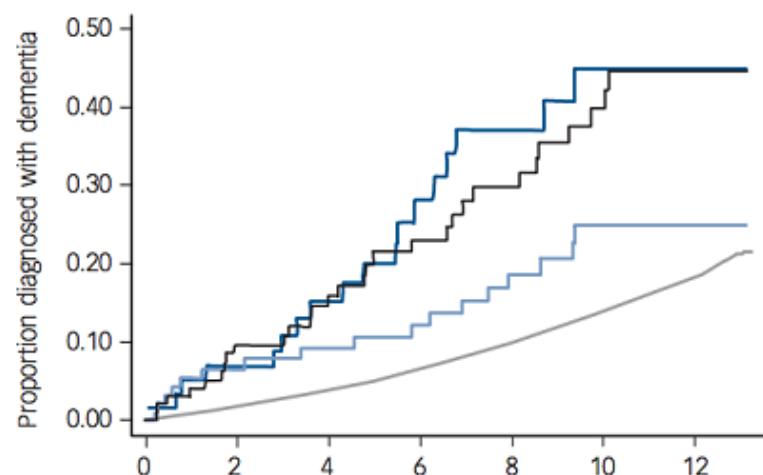
Question **19**

Not answered

Marked out of 1.00







	Years of follow-up						
Number at risk	0	2	4	6	8	10	12
Not bipolar	37 512	34 762	31 716	28 517	25 382	22 165	7028
Bipolar: <5 years	61	53	43	34	30	21	7
Bipolar: 5–14 years	95	81	73	65	55	42	19
Bipolar: ≥15 years	100	86	76	65	55	43	8

Fig. 2 Age-adjusted cumulative hazard of dementia according to the time lived with the diagnosis of bipolar disorder.

Not bipolar (grey line – reference), <5 years (dark blue line; hazard ratio (HR) 3.23, 95% CI 2.03–5.14), 5–14 years (pale blue line; HR = 1.71, 95% CI 1.06–2.76), ≥15 years (black; HR = 3.09, 95% CI 2.16–4.43). The respective age-adjusted sub-hazard ratios of dementia were 2.69 (95% CI 1.62–4.48), 1.41 (95% CI 0.86–2.31) and 2.68 (95% CI 1.81–3.95), with death used as a competing risk.

Table 2 Principal causes of death in community-representative sample of older men and among those with bipolar disorder over a follow-up period of 13 years

	Total sample (<i>n</i> = 37768) <i>n</i> (%)	Bipolar disorder (<i>n</i> = 256) <i>n</i> (%)	Mortality sub-hazard ratio for bipolar disorder ^a	95% CI
Alive	20 433 (54.1)	104 (40.6)	1	Reference
Causes of death				
Suicide	58 (0.1)	5 (1.9)	13.43	5.35–33.73
Accident	260 (0.7)	5 (1.9)	2.78	1.14–6.75
Infection	314 (0.8)	3 (1.2)	1.35	0.43–4.20

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Cancer	4484 (11.9)	41 (18.0)	1.33	0.97-1.81
Diseases of the heart	4273 (11.3)	26 (10.2)	0.89	0.58-1.26
Diseases of the brain	3758 (9.9)	38 (17.1)	0.12	0.07-0.19
Infectious diseases	3021 (7.9)	16 (7.3)	1.1	0.72-1.67
Respiratory diseases	1636 (4.3)	10 (3.9)	0.86	0.46-1.60
Liver and digestive diseases	568 (1.5)	8 (3.1)	2.02	1.00-4.06
Kidney diseases	421 (1.1)	4 (1.6)	1.35	0.50-3.63
Other causes	Men's S520 (1.4)	5 (1.9)	1.36	0.56-3.29

Select one:

- ☐ Negative likelihood ratio
- ☐ a. Mortality sub-hazard ratio derived from an age-adjusted competing risk regression model. Results in table are statistically significant.
- ☐ Diagnostic specificity
- ☐ Positive predictive value
- ☐ Negative predictive value
- ☐ Positive likelihood ratio

Your answer is incorrect.

LRs indicate by what magnitude a given diagnostic test will increase or decrease the pretest probability of the disease. Fagan's nomogram can be used for this purpose. Excerpt from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3263223/>

The correct answer is: Positive likelihood ratio

Question 20

Not answered

Marked out of 1.00

A study finds the difference in BDI scores between treated and untreated groups to be 1.06, $p < 0.001$, 95% confidence intervals 1.05-1.07. Which of the following most likely contributed to such a significant result?

Select one:

- ☐ Failure of allocation concealment
- ☐ Confounding bias
- ☐ Large sample size
- ☐ Statistical error
- ☐ Outliers

Your answer is incorrect.

The precision reported here suggests that the study was done in a large (or a highly representative) sample.

The correct answer is: Large sample size



Question **21**

Not answered

Marked out of 1.00

Natasha has been referred to the Adolescent Mental Health Team following an episode of self-harm in which she used a razor to superficially lacerate her thighs. On further exploration, she tells you that her boyfriend had recently broken up with her, and she has been feeling terrible about herself. She describes to you her struggles with intense, fluctuating emotions, identity confusion, impulsivity and frequent suicidal ideation. This has been going on for a few years and makes it difficult for her to maintain friendships. She asks you if there are any interventions that have good evidence of helping young people with difficulties similar to her own. You recommend:

Select one:

- ☐ DBT
- ☐ Interpersonal Therapy
- ☐ Fluoxetine
- ☐ CAT
- ☐ Family Therapy

Your answer is incorrect.

Even though there is some evidence of efficacy in the treatment of features of Borderline Personality Disorder with other therapies, the strongest evidence for its use in adolescence is for Dialectical Behavioural Therapy where it is associated with an improvement in suicidality, non-suicidal self-injurious behaviour, emotional dysregulation and depression from the beginning of the therapy to the one-year follow-up (Child and Adolesc Psych and Ment Health 2011; 5:3)

The correct answer is: DBT

Question **22**

Not answered

Marked out of 1.00

Mrs Galloway's husband died of lung cancer 12 months ago. He had been severely unwell for two months prior to his death. She has come to see you and describes her struggles to you. Which of the following features is most suggestive of depression?

Select one:

- ☐ Anger and irritability
- ☐ Visual hallucinations of her husband
- ☐ Excessive guilt feelings
- ☐ Preoccupation with the deceased
- ☐ Denial

Your answer is incorrect.

An excessive guilt feeling is most suggestive of depression.

Feelings of guilt can be normal, after the death of someone close. But if they are strong and prolonged, then it is suggestive of depressive reaction. Visual hallucinations may also be part of normal grief, but if they are frequent, persistent, and disruptive to functioning then a psychotic illness should be considered.

The correct answer is: Excessive guilt feelings



Question **23**

Not answered

Marked out of 1.00

Maria is a 22-year-old woman who tells you about her impulse to turn the light switch on and off seventeen times every time she leaves a room. She says that if she doesn't do it, she fears an explosion might happen and her family will be severely injured. Maria says that given that such an explosion hasn't happened, the repetitive behaviour must be working. What defence mechanism is most associated with OCD?

Select one:

- ☐ Reciprocal inhibition
- ☐ Positive reinforcement
- ☐ Denial
- ☐ Positive punishment
- ☐ Negative reinforcement

Your answer is incorrect.

Described by Skinner, and part of operant conditioning, negative reinforcement is a behaviour that is strengthened by avoiding a potential negative outcome. In this case Maria thinks that by switching the light on and off a number of times she is preventing injury occurring to her family.

The correct answer is: Negative reinforcement

Question **24**

Not answered

Marked out of 1.00

Mrs Holloways' husband died of cardiac failure 3 months ago. She is grieving for the loss of her husband. Which of the following is a feature of normal grief reaction?

Select one:

- ☐ Inability to function
- ☐ Auditory hallucinations other than the deceased
- ☐ Pining or searching for the deceased
- ☐ Visual hallucinations other than the deceased
- ☐ Strong suicidality

Your answer is incorrect.

Pining or searching for the deceased is a feature of normal grief reaction. Features of atypical grief reaction include; Strong suicidality, Inability to function, visual and auditory hallucinations other than the deceased.

The correct answer is: Pining or searching for the deceased



Question **25**

Not answered

Marked out of 1.00

Your patient reports that her coworker has got several promotions and bonuses, but she has been working in the company longer than her. She appears resentful and is considering quitting her job. Which of the following cognitive biases does she suffer from?

Select one:

- ☐ Fallacy of fairness
- ☐ Overgeneralisation
- ☐ Jumping to conclusions
- ☐ Selective abstraction
- ☐ Fallacy of misattribution

Your answer is incorrect.

Your patient seems to be suffering from the so-called fallacy of fairness. People with this cognitive bias often firmly believe that life is always supposed to be fair, and feel resentful when it turns out not to be fair.

The correct answer is: Fallacy of fairness

Question 26

Not answered

Marked out of 1.00

Miss Free has a diagnosis of Down Syndrome and her IQ is within the normal range. Which of the following chromosomal abnormalities causing Down Syndrome is she likely to have?

Select one:

- ☐ Mosaicism
- ☐ Ring chromosome
- ☐ Isochromosome
- ☐ Trisomy 21
- ☐ Translocation

Your answer is incorrect.

Mosaicism will generally result in a less severe phenotype and, in cases where a smaller proportion of cells are trisomic, an individual may only be affected to a limited degree. The other abnormalities are associated with variably reduced IQ. Robertsonian translocations are associated with a risk of recurrence in future children, depending on if a parent has a balanced translocation themselves.

The correct answer is: Mosaicism

Question **27**

Not answered

Marked out of 1.00

Ian Jones is a 12-year-old boy with a diagnosis of ADHD. His mother is keen to know more about what could have led to him developing this condition. Which of the following is a recognised risk factor for ADHD?

Select one:

- ☐ Food additives
- ☐ Rural birth
- ☐ Refugee status
- ☐ Premature birth at 35 weeks
- ☐ Older father

Your answer is incorrect.

Prematurity is an established risk factor for ADHD, as are: family history, maternal use of alcohol, drugs or tobacco during pregnancy, prenatal or perinatal obstetric complications, severe early deprivation, and exposure to environmental toxins, such as lead.

The correct answer is: Premature birth at 35 weeks



Question **28**

Not answered

Marked out of 1.00

Dillan is 6 years old. His parents report that he has been found to have moderate learning difficulties. He is rather socially anxious, makes poor eye contact, is prone to losing his temper quite easily. On examination he has large ears and a long face. What is the most likely genetic disorder?

Select one:

- ☐ Reye's syndrome
- ☐ Fragile X Syndrome
- ☐ Asperger's syndrome
- ☐ Down's syndrome
- ☐ William's Syndrome

Your answer is incorrect.

Fragile X occurs in both genders but is more common in males than females; the FMR1 Gene is implicated, located on the X chromosome.

The correct answer is: Fragile X Syndrome



Question 29

Not answered

Marked out of 1.00

Mark and Jenny's 26-month old daughter has been observed to lose many of the motor and verbal skills she had previously developed. For example, she is now unable to walk, her hands are making repeated movements seemingly without purpose, she no longer uses sentences in her speech. She has also begun to have seizures. Which of the following condition is this clinical picture highly suggestive of?

Select one:

- ☐ Retts Syndrome
- ☐ Down's Syndrome
- ☐ Edward's Syndrome
- ☐ Patau syndrome
- ☐ Ehlers Danlos Syndrome

Your answer is incorrect.

Retts Syndrome is a rare, non-inherited genetic postnatal neurological disorder that occurs almost exclusively in girls resulting in severe mental and physical disability, including due to seizures.

The correct answer is: Retts Syndrome



Question 30

Not answered

Marked out of 1.00

Eva has a diagnosis of high functioning autism. She has a particular fascination with patterns in music and is able to perceive and describe complex reoccurring patterns. What description can be given to this particular talent?

Select one:

- ☐ Extreme male brain
- ☐ Selective abstraction
- ☐ Hypersystemization
- ☐ Operant conditioning
- ☐ Positive reinforcement

Your answer is incorrect.

Systematizing is the drive to analyse or construct systems and recognising repeating patterns in stimuli. This is a common feature of those with autistic spectrum disorders (see Philos Trans R Soc Lond B Biol Sci 2009; 364 (1522): 1377-1383)

The correct answer is: Hypersystemization

Question **31**

Not answered

Marked out of 1.00

Mr Heron is a 70-year-old man who has been admitted to hospital with acute onset confusion and agitation. He is out-of-area and there is little information available about his past history. Investigations, including blood tests, are performed to try and determine a cause for his presentation. Which of the following blood results are most likely to be responsible?

Select one:

- ☐ Haemoglobin - 11g/dL
- ☐ Urea - 6mmol/L
- ☐ Sodium - 115mmol/L
- ☐ Potassium - 2.7mmol/L
- ☐ Albumin - 32g/L

Your answer is incorrect.

The diagnosis is Delirium. Hyponatraemia is more likely to cause delirium than the other abnormalities listed. Hypokalaemia is more likely to cause fatigue, constipation and muscular abnormalities, including cardiac muscle. A urea of that level could be normal for this patient in view of the usual deterioration in renal function with age. A mild anaemia would not usually cause delirium, nor would a small reduction in albumin.

The correct answer is: Sodium - 115mmol/L

Question **32**

Not answered

Marked out of 1.00

Mr Brown is a 74-year-old man whose mental state has been deteriorating significantly over the past few weeks. The consultant is considering treating him with ECT. Which of the following conditions is an indication for ECT?

Select one:

- ☐ Hypomania
- ☐ Catatonia
- ☐ Persistent Delusional Disorder
- ☐ Acute Confusional State
- ☐ Paranoid Personality Disorder

Your answer is incorrect.

Catatonia, associated with self-neglect and poor or absent oral intake, can be life-threatening, and ECT is indicated in such instances to preserve life.

The correct answer is: Catatonia

Question **33**

Not answered

Marked out of 1.00

Mrs Bainbridge is a 64-year-old woman with moderate dementia whose family have contacted mental health services as they are concerned with a change in her outlook to life. Over the past month or so she seems to be demotivated, withdrawn, and she's lost the positive outlook that was the hallmark of her personality. Which part of the brain is most likely to be affected?

Select one:

- ☐ Frontal lobe
- ☐ Occipital Lobe
- ☐ Brain stem
- ☐ Temporal lobe
- ☐ Cerebellum

Your answer is incorrect.

The frontal lobe is responsible for shaping our personality, and changes relating to dementia may result in a change in the personality as described in this vignette.

The correct answer is: Frontal lobe



Question **34**

Not answered

Marked out of 1.00

Mr Colgan has dementia of Alzheimer's type and keen on taking medications. He has preexisting cardiac problems. The drug of choice is;

Select one:

- ☐ Aspirin
- ☐ Galantamine
- ☐ Donepezil
- ☐ Gingko biloba
- ☐ Rivastigmine

Your answer is incorrect.

Among Cholinesterase inhibitors, Rivastigmine is the safest to prescribe for patients with pre-existing cardiac problems mainly due to its lack of interaction with cardiac drugs.

The correct answer is: Rivastigmine



Question 35

Not answered

Marked out of 1.00

Mr Brown is a 35-year-old man who has been admitted to a general surgical ward with an infected groin abscess. During the night one of the staff nurses observes him removing a needle from his bag, inserting it into the abscess area and then putting it into an area of unaffected skin on his opposite leg. When confronted, he has no ulterior motive for this act. Which of the following is the most likely diagnosis?

Select one:

- ☐ Factitious disorder
- ☐ Conversion disorder
- ☐ Malingering
- ☐ Hypochondriasis
- ☐ Somatisation disorder

Your answer is incorrect.

This is most likely to be a factitious disorder, as it appears that the patient is inducing an illness for an unclear gain. As there is no clear secondary gain then it is less likely to represent malingering. Hypochondriasis involves overvalued ideation about having a serious illness, whereas conversion and somatisation disorder involve patients presenting with neurological or somatic complaints, often secondary to underlying anxiety or psychological conflict.

The correct answer is: Factitious disorder

Question **36**

Not answered

Marked out of 1.00

Thomas Bailey is a 56-year-old man who had been working with the Crisis Team for a few weeks. You receive a telephone call from the Police letting you know about a tragic occurrence: Mr Bailey has killed his wife and then proceeded to take his own life. Which of the following conditions is most associated with murder-suicide?

Select one:

- ☐ Depression
- ☐ Schizophrenia
- ☐ Heroid dependence
- ☐ OCD
- ☐ Personality Disorder

Your answer is incorrect.

Depression, especially in men, is the mental illness most closely linked with murder-suicide (J of Crim Justice 27 (4):361-370)

The correct answer is: Depression



Question **37**

Not answered

Marked out of 1.00

Ben Underwood is a 34-year-old man whom the Police have contacted you about. They have received several reports from a 28-year-old woman stating that Mr Underwood is following her in public spaces, and waiting outside her door. They used to be in a relationship together, which ended some months ago, and even though she's asked him to leave her alone, he keeps pursuing her. Which if the following diagnosis is most commonly made in stalkers?

Select one:

- ☐ Erotomania
- ☐ Persistent Delusional Disorder
- ☐ Antisocial Personality Disorder
- ☐ Delusional Jealousy
- ☐ Schizophrenia

Your answer is incorrect.

Most stalkers are not psychotic and have features of Axis II disorders, in particular Antisocial PD (Mullen et al 2000, Stalkers and Their Victims, Cambridge University Press ISBN 0-521-66950-2)

The correct answer is: Antisocial Personality Disorder

Question **38**

Not answered

Marked out of 1.00

Alan Davies is a 11-year-old who has just been diagnosed with ADHD and Tourette's in a local CAMHS clinic. Which of the following drugs should be prescribed for him in the first instance?

Select one:

- ☐ Alpha 2 adrenergic agonist
- ☐ SNRI
- ☐ TCA
- ☐ SSRI
- ☐ Methylphenidate

Your answer is incorrect.

It is generally recommended that alpha 2 adrenergic agonists, such as Clonidine, are used first-line in patients with ADHD and Tourette's. There is some evidence that stimulants, such as Methylphenidate, can increase the frequency and severity of tics. However, further evidence suggests that Methylphenidate can be well-tolerated in many patients, although Dexamphetamine is thought to significantly worsen tics and should be avoided.

The correct answer is: Alpha 2 adrenergic agonist

Question 39

Not answered

Marked out of 1.00

Matthew Twist is a 17-year-old boy with a diagnosis of conduct disorder. He attends your clinic with his mother but just before you call them in, she asks to speak to you in private for a few moments. What is the best course of action?

Select one:

- ☐ Speak to the mother and then continue with the consultation
- ☐ Insist that she speaks in front of her son
- ☐ Deny the request and continue with the consultation
- ☐ Ask permission of the boy and speak to the mother if permitted
- ☐ Ask another member of staff to sit in on the consultation

Your answer is incorrect.

Children older than 16 years of age can generally be assumed to have capacity to make decisions about their treatment and care and should be consulted regarding these matters. However, until they are 18 years, their parents can still consent on their behalf, although resolving situations like this may involve the courts. Realistically, most 16 and 17 year olds will consent to such a request.

The correct answer is: Ask permission of the boy and speak to the mother if permitted

Question 40

Not answered

Marked out of 1.00

Tommy is 7 years old. His mother describes that life at home with him is very challenging: he is frequently not paying attention to her, he moves around the house very quickly, knocking things over, barging into her room without knocking. His teacher reports that at school he frequently interrupts her, cannot sit still in his chair and is so distracted he rarely manages to finish a piece of work. You consider starting him on medication for ADHD, but in taking a history you also discover he has an autoimmune liver disorder. Which of the following drugs is contraindicated in patients with ADHD and liver disease?

Select one:

- ☐ Methylphenidate
- ☐ Atomoxetine
- ☐ Fluoxetine
- ☐ Imipramine
- ☐ Dexamphetamine

Your answer is incorrect.

Atomoxetine is extensively metabolised in the liver by cytochrome P450 and associated with liver injury; its use is therefore contraindicated in patients with liver disease.

The correct answer is: Atomoxetine

Question **41**

Not answered

Marked out of 1.00

You are working with a family in which one of the children has experienced sexual abuse. Which disorder is she most likely to develop in adulthood?

Select one:

- ☐ Depression
- ☐ Autistic Spectrum Disorder
- ☐ Personality Disorder
- ☐ ADHD
- ☐ OCD

Your answer is incorrect.

Personality Disorder is the adulthood diagnostic grouping with the highest relative risk in children who have experienced childhood sexual abuse. (BJPsych 2004, 184:416-421)

The correct answer is: Personality Disorder



Question 42

Not answered

Marked out of 1.00

Stella is a 15-year-old girl who has been losing weight over the preceding 9 months. Her mother reports that she counts the number of calories in her food, avoids eating with the rest of the family, and has taken to running several kilometres every day. What are the most important features of an initial assessment in clinic?

Select one:

- ☐ IQ measurement
- ☐ Appraisal of school performance
- ☐ Family history of eating disorders
- ☐ Measurement of BMI
- ☐ Exploration of family dynamics

Your answer is incorrect.

Measurement of weight and height, so as to work out the BMI, is crucial in the initially assessment as it will help to steer the direction of subsequent consultations, investigations and interventions. A BMI below 17.5 is a feature of anorexia nervosa.

The correct answer is: Measurement of BMI



Question 43

Not answered

Marked out of 1.00

Natasha is a 19-year-old woman with a BMI of 15.8; she has been restricting her food intake for the past six months. She exercises daily in an effort to lose more weight. She believes she is overweight, despite evidence to the contrary. She does not wish to engage in talking therapy. You consider starting her on medication. Which of these would be the best initial option?

Select one:

- ☐ Sertraline
- ☐ None of the above
- ☐ Citalopram
- ☐ Fluoxetine
- ☐ Quetiapine

Your answer is incorrect.

Pharmacological medication is not indicated as the sole treatment of anorexia nervosa. It may be used to treat co-morbid mental illness, such as depression, once the patient's weight has stabilised. Meta-analyses in the use of antipsychotics have failed to show efficacy for body weight and other anorexia-related outcomes (J Clin Psychiatry 2012; 73(6):e757-66).

The correct answer is: None of the above

Question 44

Not answered

Marked out of 1.00

Mr Evans is a 42-year-old man with a diagnosis of treatment resistant schizophrenia. He is currently taking Clozapine and his serum levels are near the top limit of the usual therapeutic range. He continues to experience anhedonia and negative symptoms which interfere with his daily life. Which of the following medications could be used to augment his treatment?

Select one:

- ☐ Lithium
- ☐ Valproate
- ☐ Quetiapine
- ☐ Venlafaxine
- ☐ Mirtazapine

Your answer is incorrect.

Mirtazapine has been shown to have some beneficial effects in terms of anhedonia and negative symptoms in patients with treatment resistant schizophrenia taking Clozapine. Lithium can be used to increase white blood cell counts and may have benefits in patients with schizoaffective disorder due to mood stabilisation but is not useful in patients with schizophrenia. Valproate is often used prophylactically for seizures in patients on high doses of Clozapine; however, it does not lead to any improvement in psychotic symptoms. There is no good evidence for Quetiapine or Venlafaxine providing benefits in patients on Clozapine. Amisulpride is often used as augmentation as well.

The correct answer is: Mirtazapine

Question 45

Not answered

Marked out of 1.00

Miss Gordon has attended her local drug and alcohol clinic as she wants to stop smoking heroin. Which of the following drugs could be prescribed to her as part of an opioid detoxification?

Select one:

- ☐ Buprenorphine
- ☐ Baclofen
- ☐ Naloxone
- ☐ Buspirone
- ☐ Bupropion

Your answer is incorrect.

Buprenorphine (Subutex) is used as an opiate replacement treatment for patient's dependent upon opioids, such as heroin. It is partial opioid agonist and prevents withdrawals and any euphoriant effects if heroin is used concurrently. Baclofen is being researched to determine whether it may be useful in alcohol dependence. Naloxone is a short-acting opioid antagonist and would induce an acute withdrawal state in a patient with an opiate dependence. Buspirone and Bupropion have no role in opiate detoxification.

The correct answer is: Buprenorphine

Question 46

Not answered

Marked out of 1.00

Mrs Keeley is a 28-year-old woman who is 12 weeks' post-partum. Her husband brings her to the GP as he is concerned that she is not sleeping well and unable to enjoy things. She is fatigued, irritable and washes her baby's feeding bottle several times each day, worried that it is contaminated. Which of the following is the most likely diagnosis?

Select one:

- ☐ OCD
- ☐ Generalised anxiety disorder
- ☐ Germ Phobia
- ☐ Post-natal depression
- ☐ Puerperal psychosis

Your answer is incorrect.

The primary diagnosis is Postnatal depression. This presentation is most likely to represent an episode of post-natal depression as the patient has mood changes, in terms of irritability, and biological symptoms in the form of fatigue. Her persistent washing of bottles is likely to represent co-morbid anxiety, which has an obsessive flavour. Discrete OCD does not usually present with mood changes or fatigue.

The correct answer is: Post-natal depression

Question **47**

Not answered

Marked out of 1.00

Miss Maddison is a 24-year-old woman with a diagnosis of anorexia nervosa. She attends your clinic after having blood tests performed at her GP surgery. You have an opportunity to review these results. Which of the following blood abnormalities would lead to you making a referral to the medical team?

Select one:

- ☐ Haemoglobin - 10g/dL
- ☐ Potassium- 2.4 mmol/L
- ☐ Albumin - 30g/Dl
- ☐ Sodium - 150mmol/L
- ☐ Urea - 1.6mmol/L

Your answer is incorrect.

Hypokalaemia is potentially life-threatening due to the risk of cardiac arrhythmias. A low haemoglobin, albumin and urea are relatively common in patients with anorexia nervosa due to poor dietary and low body weight. Hypernatraemia can represent dehydration but this would not necessarily require admission provided a patient would increase their fluid intake.

The correct answer is: Potassium- 2.4 mmol/L

Question **48**

Not answered

Marked out of 1.00

Mr Walsh is a 45-year-old man who is about to undergo ECT. He is on a medication, which requires the dose of ECT to be reduced. Which of the following drugs is he likely to be taking?

Select one:

- ☐ Sertraline
- ☐ Citalopram
- ☐ Risperidone
- ☐ Olanzapine
- ☐ Lithium

Your answer is incorrect.

Lithium is associated with an increased risk of delirium, prolonged seizures, toxic lithium levels, and prolonged neuromuscular blockade. A combination of ECT and lithium is not absolutely contraindicated but the increased risks must be borne in mind when considering the relative advantages and disadvantages. This would generally restrict concurrent use to emergencies. Lithium can be held the day before, maintained at the lowest therapeutic dose or changed to an alternative medication.

The correct answer is: Lithium

Question 49

Not answered

Marked out of 1.00

Mr Price is a 40-year-old man who is being treated for a first episode of mania. He has been started on Olanzapine to treat this. How long after he recovers should he continue to take an antipsychotic medication?

Select one:

- ☐ 2 years
- ☐ 3 months
- ☐ 1 year
- ☐ 1 month
- ☐ 6 months

Your answer is incorrect.

There is reasonable evidence that continuing antipsychotic medication for 6 months after the resolution of an episode of mania reduces relapse rates compared to stopping it abruptly. However, there is no strong evidence for continuing for longer than 6 months.

The correct answer is: 6 months



Question 50

Not answered

Marked out of 1.00

Puerperal psychosis is an uncommon but serious mental health problem that tends to arise in the first days and weeks after a woman gives birth. How frequently does it occur?

Select one:

- ☐ 0.5-0.9 per 1000 births
- ☐ 5-8 per 1000 births
- ☐ 10-12 per 1000 births
- ☐ 3-4 per 1000 births
- ☐ 1- 2 per 1000 births

Your answer is incorrect.

Puerperal psychosis occurs in around 1-2 per 1000 births

The correct answer is: 1- 2 per 1000 births



Question 51

Not answered

Marked out of 1.00

Mr Richardson is a 35-year-old man with a diagnosis of hypochondriasis. He has started to respond to treatment for this but is still struggling with his daily functioning and is wondering if there is some other problem occurring. Which of the following co-morbid mental health conditions is he most likely to be experiencing?

Select one:

- ☐ Somatisation disorder
- ☐ Generalised anxiety disorder
- ☐ Social phobia
- ☐ Agoraphobia with panic disorder
- ☐ Depression

Your answer is incorrect.

Patients with hypochondriasis have a high rate of psychiatric comorbidity. In one outpatient clinical series, 88% of patients with hypochondriasis had one or more concurrent psychiatric disorders, the most common being generalized anxiety disorder (71%), dysthymic disorder (45.2%), major depression (42.9%), somatization disorder (21.4%), and panic disorder (16.7%). These patients are 3 times more likely to have a personality disorder than the general population. Arch Gen Psychiatry. Feb 1992;49(2):101-8.

The correct answer is: Generalised anxiety disorder

Question **52**

Not answered

Marked out of 1.00

Miss Saunders is a 21-year-old woman with a diagnosis of anorexia nervosa. She has asked about starting on some medication in addition to having psychotherapeutic interventions. Which of the following medications should be avoided in her case?

Select one:

- ☐ Citalopram
- ☐ Sertraline
- ☐ Venlafaxine
- ☐ Mirtazapine
- ☐ Fluoxetine

Your answer is incorrect.

SSRIs are generally used in anorexia nervosa if pharmacological treatment is being considered. Venlafaxine has also been studied in clinical trials. While concerns have been raised about QTc prolongation with high dose Citalopram this can be managed via monitoring ECGs. Citalopram has one of the better evidence bases for medications in anorexia nervosa. Mirtazapine can markedly increase appetite, which may be distressing for patients recovering from anorexia nervosa.

The correct answer is: Mirtazapine

Question **53**

Not answered

Marked out of 1.00

Mrs Trevor is a 28-year-old woman with a diagnosis of schizophrenia who has recently had a baby and is breastfeeding. She is well at present but is continuing with medication as she is at high risk of relapse. Which of the following medications should be avoided in her case?

Select one:

- ☐ Olanzapine
- ☐ Quetiapine
- ☐ Amisulpride
- ☐ Clozapine
- ☐ Risperidone

Your answer is incorrect.

Clozapine should be avoided in breastfeeding mothers as it is transferred into the breast milk and there have been recorded cases of neonatal agranulocytosis as a result.

The correct answer is: Clozapine



Question **54**

Not answered

Marked out of 1.00

Mrs Ula is 40-year-old woman with a diagnosis of bipolar affective disorder. She has been without medication for around 3 years and has recently developed a depressive episode. She is expressing significant suicidal ideation and her family have asked which medications are best for preventing suicide. Which of the following medications have been shown to reduce rates of suicide in patients with bipolar affective disorder?

Select one:

- ☐ Sodium Valproate
- ☐ Olanzapine
- ☐ Lithium
- ☐ Venlafaxine
- ☐ Clozapine

Your answer is incorrect.

Lithium has been shown to reduce the incidence of suicide in patients with bipolar affective disorder. Clozapine has been shown to reduce the incidence of suicide in patients with treatment resistant schizophrenia. The other medications listed have not been shown to specifically reduce the rate of suicide.

The correct answer is: Lithium

Question 55

Not answered

Marked out of 1.00

The suicide rate for women worldwide is significantly lower than for men. Which of the following suicide methods are most commonly used by women in the UK?

Select one:

- ☐ Jumping from height
- ☐ Cutting
- ☐ Overdose
- ☐ Drowning
- ☐ Hanging

Your answer is incorrect.

Women in the UK are most likely to complete suicide via overdose than other methods. Men are more likely to use violent methods, such as hanging or jumping. This can be different in other areas.

The correct answer is: Overdose

Question **56**

Not answered

Marked out of 1.00

Miss Viola is a 19-year-old woman who has been brought to the local Emergency Department following an overdose. Which of the following is she most likely to have taken?

Select one:

- ☐ Ibuprofen
- ☐ Paracetamol
- ☐ Phenytoin
- ☐ Heroin
- ☐ Cocaine

Your answer is incorrect.

Paracetamol is the most common substance taken in overdose worldwide.

The correct answer is: Paracetamol



Question **57**

Not answered

Marked out of 1.00

Tom Bomball has had two admissions to a psychiatric unit with features of mania with psychosis. In between these episodes he's also had periods of severe depression, for which he required Home Treatment Team intervention. Mr Bomball also has an inherited renal disorder for which he requires a renal transplant. What is the best treatment option for him?

Select one:

- ☐ Sodium Valproate
- ☐ Vigabatrin
- ☐ Lamotrigine
- ☐ Lithium
- ☐ Carbamazepine

Your answer is incorrect.

The features of renal impairment include reduced protein binding, and accumulation of metabolites. Lithium in particular should be avoided in severe renal impairment, given it is renally excreted and its accumulation can lead to Lithium toxicity. Carbamazepine and Lamotrigine have both been associated with adverse events in patients with renal impairment, and should be used cautiously and at lowered doses than usual. Studies conducted on Sodium Valproate and renal function have not demonstrated clinically significant adverse effects (Ren Fail 2006;28(7):593-7)

The correct answer is: Sodium Valproate

Question 58

Not answered

Marked out of 1.00

Kieran Todd is a 53-year-old man of Kenyan origin who presents to his GP with gradual onset and progressive head tremor, difficulties with coordination and pronouncing of some words, and behavioural changes (eg increased impulsivity and emotional incongruence). His GP does a routine set of bloods, and discovers Mr Todd is HIV positive. He refers Mr Todd for an MRI of the brain. What is the most likely brain change associated with HIV?

Select one:

- ☐ Symmetrical hyperintensity of the pulvinar nuclei of the thalamus
- ☐ Hydrocephalus
- ☐ Progressive multifocal leukoencephalopathy
- ☐ Atrophy of angular gyrus
- ☐ Reversal of asymmetry of planum temporale

Your answer is incorrect.

Progressive multifocal leukoencephalopathy (PML) is associated with HIV; HIV infection accounts for almost 85% of the total cases of PML. It is currently one of the AIDS-defining illnesses in HIV-infected patients (Neuroscientist 2010 16(3):308-23)

The correct answer is: Progressive multifocal leukoencephalopathy

Question 59

Not answered

Marked out of 1.00

Mrs Dundheim comes to see you in clinic, accompanied by her adult daughter. You notice that there is some disturbance in her movements. Her daughter tells you that her mother's sister had Parkinson's Disease, and she's worried her mother might also have it. Which of the following physical signs is most indicative of Parkinson's disease?

Select one:

- ☐ Pill-rolling tremor
- ☐ Limb apraxia
- ☐ Tachykinesia
- ☐ Increased facial expressions
- ☐ Broad-based gait

Your answer is incorrect.

A pill-rolling tremor, occurring at rest, is suggestive of Parkinson's disease, alongside bradykinesia, rigidity (both sides of the body, may start as unilateral rigidity), and mask-like facial appearance due to reduced facial expressions.

The correct answer is: Pill-rolling tremor

Question 60

Not answered

Marked out of 1.00

The mental health liaison nurse has assessed a person presenting with suicidal ideas in the emergency department and asks you conduct a review. She is concerned that this person is at high risk. You look over her notes. What description of the patient is linked with the highest risk of suicide?

Select one:

- ☐ 55-year-old man, recently widowed, unemployed
- ☐ 24-year-old man, unemployed, single
- ☐ 32-year-old woman, married, subject to domestic violence
- ☐ 85-year-old man, married, wife is unwell
- ☐ 16-year-old woman, recently split up from boyfriend

Your answer is incorrect.

Middle age, male gender, unemployment, with recent major loss are all significant factors in contributing to the risk of suicide. This is made even higher by alcohol and/or drug misuse.

The correct answer is: 55-year-old man, recently widowed, unemployed

Question **61**

Not answered

Marked out of 1.00

Ben White is a 22-year-old student who attends A&E complaining of feeling out of touch with his body. He's not been able to sleep all night, he has stomach cramps, and on observation he's sweating profusely. He is worried it might be related to drug use. Which of the following clinical states is the best explanation?

Select one:

- ☐ Amphetamine intoxication
- ☐ Heroin withdrawal
- ☐ Inhalant misuse
- ☐ Cocaine withdrawal
- ☐ Cannabis intoxication

Your answer is incorrect.

These are typical features of withdrawal from opioids, also accompanied by piloerection, muscle spasms, and tremors.

The correct answer is: Heroin withdrawal



Question **62**

Not answered

Marked out of 1.00

You are due to visit a 85 year old woman who lives alone, and who has for some months been complaining about her neighbour spying on her, listening to her movements through the walls, tapping on the walls when she's trying to sleep. Which if the following feature is most likely to be seen in persistent delusional disorder?

Select one:

- ☐ Ideas of reference
- ☐ Non-bizarre delusions
- ☐ Passivity phenomena
- ☐ Tactile hallucinations
- ☐ Auditory hallucinations

Your answer is incorrect.

Persistent delusional disorder is often characterized by life long persistent non-bizarre delusions. The others are more suggestive of a broader psychotic illness, including schizophrenia.

The correct answer is: Non-bizarre delusions



Question **63**

Not answered

Marked out of 1.00

June Watson is a 47-year-old woman who, for the past month and since the loss of her job, has been feeling demotivated and persistently low in mood. She describes difficulty in getting to sleep, waking at 4am most mornings, reduced interest in food, poor concentration and low sexual energy. What's the diagnosis?

Select one:

- ☐ Moderate depression
- ☐ Mild depression
- ☐ Severe depression with psychosis
- ☐ Severe depression
- ☐ Manic depression

Your answer is incorrect.

Moderate depression includes the two core features (low mood and anhedonia) with four or more related symptoms.

The correct answer is: Moderate depression



Question 64

Not answered

Marked out of 1.00

Jason is a 20-year-old man who first presented to mental health services with third person auditory hallucinations (running commentary) and delusions of persecution, which lasted eight months before a more effective approach helped them to diminish. He has now been relatively free of psychotic symptoms for the past nine months, except for occasional paranoia; his mother tells you that for the past three weeks she's observed him sleeping much more than usual, he's eating little and seems to have lost motivation. You meet with him and he tells you he's not interested in much, he's finding it difficult to concentrate, he's no longer enjoying the usual computer games that he used to look forward to playing, and generally feeling quite hopeless. What the most likely diagnosis?

Select one:

- ☐ Severe depressive illness
- ☐ Moderate depressive illness
- ☐ Simple schizophrenia
- ☐ Post-schizophrenic depression
- ☐ Schizotypal disorder

Your answer is incorrect.

The diagnosis of Post schizophrenic depression is made if the patient has met general criteria for schizophrenia within the past 12 months, with some schizophrenic symptoms remaining, and depressive symptoms having become prominent and distressing for at least 2 weeks, fulfilling criteria for a depressive episode.

The correct answer is: Post-schizophrenic depression

Question 65

Not answered

Marked out of 1.00

Rebecca has been struggling with symptoms of a moderate depressive illness for a few months. Her GP commenced her on Sertraline, and she's been taking the therapeutic dose for the past two months with little improvement in her symptoms. Which of the following is the next best course of action?

Select one:

- ☐ Start Lithium
- ☐ Stop all antidepressant medication and review
- ☐ Add an atypical antipsychotic
- ☐ Switch to a different class of antidepressants
- ☐ Switch to another SSRI

Your answer is incorrect.

Guidelines suggest switching to another SSRI. There should be a sequential trial of two SSRIs at adequate dose, for sufficient length of time, before alternatives are considered.

The correct answer is: Switch to another SSRI

Question 66

Not answered

Marked out of 1.00

Mrs Evergreen has suffered a stroke affecting the dominant frontal lobe. You are asked to review her mental state and you find that she is having difficulties in communicating verbally with you, although she seems to understand what you are saying. What type of speech problem is most likely to be present given the location of the stroke?

Select one:

- ☐ Primary Progressive aphasia
- ☐ Expressive aphasia
- ☐ Mixed non-fluent aphasia
- ☐ Anomic aphasia
- ☐ Wernicke's aphasia

Your answer is incorrect.

A dominant frontal lobe stroke affects a person's ability to produce fluent speech; language comprehension is often retained.

The correct answer is: Expressive aphasia



Question **67**

Not answered

Marked out of 1.00

Valerie has had a first episode of depression. Her brother accompanies her to the clinic. He has a friend who has had several episodes of depression, and wishes to know what the chances are for Valerie to go on to experience another episode of depression.

Select one:

- ☐ 5%
- ☐ 50%
- ☐ 10%
- ☐ 20%
- ☐ 80%

Your answer is incorrect.

50% of those who've had a first episode will go on to have another, and approximately 80% of those with a history of two episodes will have another recurrence (American Psychiatric Association 2000).

The correct answer is: 50%

Question 68

Not answered

Marked out of 1.00

Mr Jenkins has features of mild cognitive impairment. His daughter is anxious to know what the year-on-year chances are of this progressing to Alzheimer's dementia.

Select one:

- ☐ 4.6%
- ☐ 80.6%
- ☐ 45.6%
- ☐ 9.6%
- ☐ 1.6%

Your answer is incorrect.

The annual adjusted conversion rate for mild cognitive impairment (MCI) to Alzheimer's dementia is 9.6%. (Acta Psychiatr Scand 2009; 119(4):252-65

The correct answer is: 9.6%



Question 69

Not answered

Marked out of 1.00

Mr Kahn is a 32-year-old man with epilepsy, who has recently experienced job loss, and has had to relocate to a new area. Over the past month he's been experiencing early morning waking, low motivation, loss of interest in his usual hobbies, and feelings of worthlessness and hopelessness. He wishes to start a medication with antidepressant effect, but is worried about what effect it might have on his epilepsy. Which medication would you recommend?

Select one:

- ☐ Bupropion
- ☐ Lithium
- ☐ Clomipramine
- ☐ Venlafaxine
- ☐ Sertraline

Your answer is incorrect.

SSRIs are considered first-line antidepressant option for patients with epilepsy. Tricyclics should be avoided, and also the dopamine and noradrenaline-specific reuptake inhibitor Bupropion.

The correct answer is: Sertraline



Question 70

Not answered

Marked out of 1.00

Mrs Galloway has been doing some internet research to see if she can find a treatment that might help her son who has longstanding mental health difficulties. She finds some positive literature relating to Deep Brain Stimulation and asks you about it. For what conditions has Deep Brain Stimulation(DBS) shown to have therapeutic potential?

Select one:

- ☐ Persistent Delusional Disorder
- ☐ OCD
- ☐ ADHD
- ☐ Borderline Personality Disorder
- ☐ Anorexia Nervosa

Your answer is incorrect.

DBS had shown to have therapeutic potential in patients with OCD. As well as OCD, there is evidence of therapeutic effects for treatment-resistant major depressive disorder, Parkinson's Disease among others.

The correct answer is: OCD



Question 71

Not answered

Marked out of 1.00

Ted is a 36-year-old man with treatment-resistant schizophrenia who also has benign ethnic neutropenia. He has been on Clozapine for six months, and has experienced significant clinical improvement. However, his white cell count has been lowered by the use of Clozapine, and the latest neutrophil reading was just under 1000/cubic mm. What adjunct could be considered in order to keep Ted on Clozapine and reverse the Clozapine-induced neutropenia?

Select one:

- ☐ Sodium Valproate
- ☐ Olanzapine
- ☐ Amitriptyline
- ☐ Lithium
- ☐ Methyphenidate

Your answer is incorrect.

Lithium increases the white blood cell and neutrophil count, both acutely and chronically, and may be used for patients with Clozapine-induced neutropenia (Progress in Neur and Psychiatry 2016; 20(3):13-15)

The correct answer is: Lithium

Question 72

Not answered

Marked out of 1.00

Miss Isbell is a 27-year-old woman with a diagnosis of anorexia nervosa. She is currently an informal patient on a ward. She has a BMI of 15kg/m². In the evening she reports sudden-onset shortness-of-breath and chest pain to the nursing staff. Which of the following is the most likely diagnosis?

Select one:

- ☐ Pulmonary embolism
- ☐ Myocardial infarction
- ☐ Congestive heart failure
- ☐ First degree heart block
- ☐ Pneumothorax

Your answer is incorrect.

Even relative tachycardia can precipitate congestive cardiac failure in patients with anorexia nervosa and low body weight. They will often have lost cardiac muscle and will have a relatively low heart rate. If this increases, the weakened heart cannot manage the cardiac load, leading to pulmonary oedema and chest pain due to myocardial insufficiency. First degree heart block is not usually associated with heart failure. And there is nothing specific in the history to suggest pulmonary embolism, myocardial infarction or pneumothorax.

The correct answer is: Congestive heart failure

Question 73

Not answered

Marked out of 1.00

Mrs Arnold is a 32-year-old woman who has a known diagnosis of bipolar affective disorder. She is 39 weeks pregnant and has been referred to the local Perinatal clinic for advice around her lithium. She is currently taking Priadel (Lithium carbonate) at a dose of 600mg daily and her most recent lithium level was 0.75mmol/L. Her bipolar disorder is well controlled and she has not had a relapse for the past 2 years. Which of the following is the most appropriate plan of action?

Select one:

- ☐ Make no changes and continue usual follow-up schedule
- ☐ Increase dose at delivery to 800mg and recheck lithium level after 1 week
- ☐ Reduce lithium dose to 400mg
- ☐ Recheck lithium level at 5 weeks post-partum and review results at clinic one week later
- ☐ Stop lithium until after delivery

Your answer is incorrect.

During pregnancy GFR and plasma volume increase by around 50% and 40% respectively. These decrease again rapidly after delivery. A higher lithium dose is often necessary during the third trimester as a result. However, these changes rapidly reverse following delivery, which could pose a high risk of toxicity in the mother. But stopping it abruptly risks a relapse, particularly in a mother with a history of bipolar affective disorder, who could have a 50% chance of becoming acutely ill following delivery. As the patient's lithium level is well within the recommended maintenance range (0.4-1.0mmol/L) then it is reasonable to reduce the dose to 400mg prior to delivery. Lithium should be stopped when labour starts and the mother's levels checked 12 hours after her last dose. It can then be restarted following delivery and the levels checked regularly. It is advised that women do not breastfeed when taking lithium as it is transferred into the milk in significant concentrations.

The correct answer is: Reduce lithium dose to 400mg

Question 74

Not answered

Marked out of 3.00

Drug overdose

Please identify the drug that was most likely given in the following scenarios.

1. Miss Carlson is a 32-year-old woman who has brought into the Emergency Department comatose and unresponsive. Ten minutes after she is given a drug she jumps up off the bed and storms out of the department. ✖
2. Mr Davies is a 19 year old man who has been brought to the Emergency Department by friends after consuming 56 Paracetamol tablets following the end of a relationship. He is admitted and given an infusion. ✖
3. Mrs Edwards is a 40-year-old woman who has been admitted for an appendectomy. On her return from theatre she is noted to have a spontaneous respiratory rate of 6 breaths per minute. ✖

Explanation: 1) An abrupt return to consciousness following administration of a medication, particularly in a patient who immediately leaves the department, suggests opiate overdose and subsequent use of naloxone. 2) Parvolex (N-acetylcysteine) is the recommended treatment for paracetamol overdose. When paracetamol is taken in large quantities, a minor metabolite called N-acetyl-p-benzoquinone imine (NAPQI) accumulates within the body. It is normally conjugated by glutathione, but when taken in excess, the body's glutathione reserves are not sufficient to inactivate the toxic NAPQI. This metabolite is then free to react with key hepatic enzymes, thereby damaging liver cells. This may lead to severe liver damage and even death by acute liver failure. N-acetylcysteine acts to maintain or replenish depleted glutathione reserves in the liver and enhance non-toxic metabolism of paracetamol. 3) Flumazenil is used for patients with benzodiazepine overdose or excess. It acts via competitive inhibition at the benzodiazepine-binding site on the GABAA receptor. Benzodiazepines are often used during surgical procedures and these can cause continued respiratory depression in some patients.

Question 75

Not answered

Marked out of 3.00

Treatment of alcoholism.

Which of the following treatments would be most appropriate for each of the following patients?

1. Mr Fletcher is a 34-year-old man who is bingeing on alcohol and cocaine once or twice weekly. He has attended outpatient clinic as he is thinking about stopping. ❌
2. Miss Gaskell is a 47-year-old woman who has a history of alcohol dependence. She successfully completed a community detoxification one week ago and is engaged with a treatment programme. She is struggling with cravings. ❌
3. Mr Gosh completed alcohol detoxification programme in the hospital and discharged on a medication. He has marked nausea and vomiting when alcohol was consumed. ❌

Explanation: 1) Motivational interviewing is the most appropriate intervention to move patients from considering change to enacting and sustaining it. The stages of change are: precontemplation, contemplation, preparation, maintenance, and relapse. This patient would not benefit from medication as he is not physiologically dependent. 2) Acamprosate has an established indication in reducing cravings in newly abstinent patients who have been dependent upon alcohol. Naltrexone is being studied but is not as widely used. 3) Disulfiram induces marked nausea and flushing if alcohol is consumed due to antagonism of aldehyde dehydrogenase.

Question 76

Not answered

Marked out of 3.00

Addiction & Treatment options.

Which of the below options would be most appropriate in the following scenarios?

1. Mr Picard is 30-year-old man with a history of alcohol dependence and withdrawal seizures. He is keen to stop alcohol but is concerned about having another seizure and experiencing other withdrawal symptoms. Which medication should he receive?

✖

2. Mr Richardson is a 70-year-old man who has had long-standing issues with alcohol. His daughter has brought him to the GP as she says he keeps "telling lies" about things that have happened. ✖

3. Mr Simpson is a 50-year-old man who is known to have alcohol dependence. He is brought in by a member of the public who finds him confused and stumbling in the street. He is found to have nystagmus on examination. Which medication should he receive?

✖

Explanation: 1) Benzodiazepines, usually diazepam or chlordiazepoxide, are the mainstay of alcohol detoxification and are given in reducing doses until an individual is no longer physiologically dependent on alcohol. 2) This patient likely has Korsakoff syndrome with classical anterograde amnesia and associated confabulation. Family members and professionals can often interpret the latter as patients being untruthful. This disorder is most common in patients with long-standing alcohol dependence and poor dietary intake. It is due to a severe deficiency of thiamine, which is necessary for the metabolism of carbohydrates. This leads to microhaemorrhages in the medial dorsal nucleus, leading to memory impairment, as well as the mamillary bodies and thalamus. 3) The acute phase of Korsakoff's syndrome is Wernicke's syndrome. The two are often linked as Wernicke-Korsakoff syndrome. The acute phase classically involves a triad of confusion, ataxia and ophthalmoplegia, such as nystagmus or diplopia. However, this is only seen in 15% of cases and there must be a high index of suspicion in any patient with alcohol problems presenting in a confused state. The chronic phase can be prevented or mitigated in severity by the use of high dose intravenous thiamine, usually given as Pabrinex.

Question 77

Not answered

Marked out of 4.00

Substance misuse- signs and symptoms

Which of the following substance related problems are the following patients likely to be experiencing?

1. Mr Taylor is a 26-year-old man who presents with vivid unpleasant dreams and a marked increase in appetite.

2. Mr Underwood is a 35-year-old man who is brought to his local Emergency Department by friends. When there, he is agitated, hypertensive and an ECG shows ischaemic changes.

3. Miss Vincent is a 21-year-old woman who is brought into hospital by her friends. She is confused and her physical observations indicate that she is hyperthermic, tachycardiac and hypertensive.

4. Mr Read presents with sweating, increased hand tremors, restlessness, vomiting and irritability.

Explanation: 1) Amphetamine withdrawal is typified by: fatigue, hypersomnia, psychomotor agitation or retardation, increased appetite and vivid, unpleasant dreams. 2) Cocaine intoxication leads to alertness, feelings of well-being, euphoria, energy, competence, sociability, and sexuality. Common side effects include anxiety, increased temperature, paranoia, restlessness, and teeth grinding. Dangerous side effects include palpitations, arrhythmias, heart attack, heart failure, tremors, seizures, stroke and markedly increased core temperature and renal failure. 3) MDMA (3,4-Methylenedioxymethamphetamine) is a recreational drug, which is used to induce euphoria, increased empathy and heightened sensations. Some individuals can experience adverse reactions with marked dehydration, hyperthermia, hypertension, tachycardia and agitation. 4) Symptoms of alcohol withdrawal include sweating, hand tremors, restlessness, shakiness and craving. Gastric symptoms include nausea, retching & vomiting. Behavioural symptoms include irritability and agitation. Autonomic hyperactivity such as tachycardia and hypertension are also seen.

Question 78

Not answered

Marked out of 6.00

Disorders and Neuroimaging findings

Find the best match between each neuro-imaging finding and its corresponding disorder;

1. Reduction in cerebellar vermis and enlargement of the 4th ventricle. ✖
2. Enlarged lateral ventricles, reduced total brain volume and reduction in grey-matter volume. (Choose 2 answers)
 ✖ ✖
3. Reversal of the normal left-larger-than- right asymmetry of planum temporale. ✖
4. Cavum septi pellucidi, white matter fibre tract abnormalities. (Choose 2 answers) ✖
 ✖

Explanation:

- 1) Fragile X Syndrome: The most important abnormalities were reduction of the cerebellar vermis and enlargement of the 4th ventricle.
- 2) Down's syndrome- common findings include total brain volume loss. Grey-matter volumes were reduced, as were the volumes of the left hippocampus and amygdala; ventricle volumes were larger. More generalized atrophy for age, mesial temporal shrinkage, and third ventricular enlargement is seen. Schizophrenia- Common findings include a decrease in brain weight and volume of the cerebral hemispheres. Enlargement of the lateral ventricles is seen (especially temporal horns). Other findings include reduced tissue volume in the thalamus, in temporolimbic structures including hippocampus, amygdala and parahippocampal gyrus.
- 3) The planum temporale, the posterior superior surface of the superior temporal gyrus, is a highly lateralized brain structure involved with language. In schizophrenic patients, a consistent reversal of the normal left-larger-than- right asymmetry of planum temporale surface area is noted. Heschl's gyrus (primary auditory cortex) showed no differences between the left and right sides.

4) Velocardio facial syndrome (VCFS)- Neuroimaging findings to date show a high incidence of cavum septi pellucidi, brain atrophy and brain asymmetry in VCFS. Abnormalities in specific white matter fiber tracts are also seen. In Schizophrenia there is white-matter reductions in parahippocampal gyrus or hippocampus and an increased incidence of a cavum septi pellucidi is noted.



Question 79

Not answered

Marked out of 4.00

Treatment of bipolar disorder

Bipolar affective disorder requires different treatments when patients come to mental health services with differing presentations and previous histories. Which of the following treatments would be most appropriate in the cases below?

1. Mr Yong is a 25-year-old man with a diagnosis of bipolar affective disorder who has developed an episode of mania. He is currently taking lithium and sertraline. ✖
2. Ms Zanetti is a 23-year-old year woman who has developed an acute episode of mania. She has a previous history of depression. She is very agitated and has required admission to hospital. ✖
3. Mrs Adams is a 45-year-old woman with bipolar disorder who has presented with an episode of severe depression. She has previously had a switch to mania when started on antidepressants. She is currently taking lithium and her levels are within therapeutic range. ✖
4. Mrs Brown is a 50-year-old woman with a diagnosis of bipolar affective disorder. She has unfortunately required transplantation for renal failure due to lithium. She is not keen to continue lithium after transplantation. ✖

1) NICE guidelines [CG185] recommend that if a patient with bipolar affective disorder develops an episode of mania and is taking an antidepressant, this should be stopped.

2) They also advise that a patient with new onset mania should be offered an antipsychotic in the first instance.

3) If a patient develops an episode of depression and is taking lithium, their serum levels should be optimised. If this is ineffective they should be offered either fluoxetine combined with olanzapine, or quetiapine, depending on their preference and previous response to treatment. In view of this patient's previous manic switch, it would be sensible to use an antipsychotic alone, or lamotrigine.

4) In patients with chronic renal failure amisulpride or sulpride are not preferred agents. Once an individual has received a transplant they can return to near-normal or normal levels of renal function. Lithium can be used in these patients, but valproate is a better choice. It is mainly metabolised in the liver and reasonably well tolerated in renal disease.

Question 80

Not answered

Marked out of 3.00

Organic psychiatry

There are various dementia types, which present with different clinical features. Which dementia type is most likely in the three following examples?

1. Mr Horner has developed cognitive deterioration since the age of 56. He and his wife are informed that he has the most common dementia type for people under the age of 65. ❌
2. Mr Illy is a 72-year-old man who is experiencing increasing difficulties with language. He struggles to speak at times and makes errors in terms of the sounds of words and spoken grammar. A childhood stutter has also returned. ❌
3. Mr Hill has developed sudden onset memory loss with ataxia. He has jerky movements in his legs during nighttime. ❌

- 1) Alzheimer's dementia is the most common dementia in people under the age of 65 years, accounting for around 30-35% of cases.
- 2) Frontotemporal dementia is associated with non-fluent aphasia. This is characterized by difficulties in the production of language such as. Hesitant, effortful speech, speech 'apraxia, stutter (including return of a childhood stutter), anomia, phonemic paraphasia (sound errors in speech e.g. 'gat' for 'cat'), and agrammatism (using the wrong tense or word order).
- 3) Creutzfeldt Jakob disease (CJD) is characterized by rapid onset memory loss, myoclonic jerks (hypnic jerks) and ataxia (due to cerebellar atrophy).

Question 81

Not answered

Marked out of 3.00

Diagnosis of neurotic disorder

Which anxiety disorder do the following examples represent?

1. Mrs Kettle is a 32-year-old woman who is 4 months postnatal. She is persistently concerned that her baby's utensils, bottles and dummies are contaminated and needs to wash and sterilise them 5 times before she allows them to be used again.

2. Mrs Longford is a 50 year old woman who has been attending her GP at least once a month for 30 years with numerous different complaints, including nausea, bowel disturbance, abdominal pain, muscular aches, headaches, palpitations, shortness of breath and urinary hesitancy. No definitive diagnosis has been found for these symptoms.

3. Mr Mitchell is a 24-year-old man who has repeatedly consulted his GP about a moderately sized mole on his right arm. He is convinced that this represents skin cancer, despite having been advised by two consultant dermatologists that there is no indication of this.

Explanation:

- 1) Obsessive compulsive disorder is characterised by recurrent worries (obsessions) and actions to address these anxieties (compulsions). In this case the patient has obsessive concerns about contamination and addresses these via excessive and compulsive washing of her baby's utensils.
- 2) Somatisation disorder usually involves an individual repeatedly presenting to health services with a wide range of somatic complaints involving numerous organ systems, often over a prolonged period of time.
- 3) Hypochondriasis involves overvalued ideation about having a serious illness. Patients can be reassured for a time by medical professionals but the anxieties usually return.

Question 82

Not answered

Marked out of 4.00

Schizophrenia (types)

Identify the most likely form of schizophrenia in the following cases:

1. Tom, a 16-year-old boy, has been increasingly withdrawn over the past six months. He doesn't show much in the way of emotion, he has little motivation to pursue anything, mostly spends time on his own; his social and academic functioning are severely impaired.



2. Amy, an 18-year-old woman, attends A&E in an extremely distressed state. She reports that Martians are planning to take over the garage on her street and that she is the only person who can act as mediator in this situation. She feels overwhelmed by this responsibility, and holds her beliefs with absolute certainty.



3. Hussein is 17 years old. His parents report that over the past year his character seems to have changed. He spends quite a lot of time on his own, playing with cars, giggling to himself. At interview his speech is disorganised and rambling. He says he believes himself to be a car, a phrase he repeats often. He has lost motivation for studying and has stopped all his social contacts.



4. Abu is a 18 year old Somalian man, who has recently moved to the UK. His family are extremely concerned about him. Yesterday he became enraged, seemingly without provocation, and began smashing furniture in his bedroom in a very energetic way. Now he is laying on his bed, and has been almost immobile for 12 hours, and yet not sleeping. His family have tried to get him out of bed. They removed the pillow from under his head, and Abu has kept his head up as though there were a pillow underneath it. They say this has happened a few times before when he was 16, and also last year.



Explanation:

1) Simple Schizophrenia: there's an insidious and progressive development of oddities of social conduct, inability to meet the demands of society; characteristic negative features of schizophrenia (e.g. blunting of affect, loss of volition) develop without being preceded by overt psychotic symptoms.

2) Paranoid schizophrenia: Characterized by prominent delusions, often persecutory in nature and auditory hallucinations. Affective and negative symptoms are less prominent.

3) Hebephrenic Schizophrenia: Affective changes are prominent (often shallow, inappropriate mood, incongruent affect with giggling). Thought disorder and loosening of associations are often prominent.

4) Catatonic Schizophrenia: there are prominent psychomotor disturbances (e.g. hyperkinesia and stupor, or automatic obedience and negativism). Postures may be maintained for long periods. Episodes of violent excitement may occur.

Question **83**

Not answered

Marked out of 3.00

Child Psychiatry - diagnosis

Which of the below options would be most appropriate in the following scenarios?

1. Samantha Norris is a 13-year-old girl with a diagnosis of ADHD with multiple motor and vocal tics. She and her family are keen on treatment. Which medication would be most appropriate for her? ✖

2. Nigel Osgood is a 16-year-old boy with a new diagnosis of childhood bipolar affective disorder. Which comorbidity is he most likely to have? ✖

3. Tommy is a 12-year-old boy who was assessed in the child development clinic and parent management training was suggested. Which childhood disorder is he most likely to have? ✖

Explanation: 1) The diagnosis is ADHD with comorbid Tourette's disorder. It is generally recommended that alpha 2 adrenergic agonists, such as Clonidine, are used first-line in patients with ADHD and Tourette's. There is some evidence that stimulants, such as Methylphenidate, can increase the frequent and severity of tics. However, further evidence suggests that Methylphenidate can be well tolerated in many patients, although Dexamphetamine is thought to significantly worsen tics and should be avoided. 2) In Paediatric populations, some studies have found that 90% of patients with bipolar affective disorder also have ADHD. Conduct disorder is also common. 3) Parent management training is the best established approach for managing children with conduct disorders and it is backed up numerous RCTs. Parents are taught to pay attention to desired behaviour. They are also taught effective techniques for handling undesirable behaviour.

Question **84**

Not answered

Marked out of 2.00

Criminal behaviour:

Which of the following mental disorders would be most likely to be associated with the below scenarios?

1. Mr Young is a 22-year-old man who has been detained by the police for staring and waving at women in a park. When arrested he was surprised and did not appear to feel he was doing anything untoward. ❌

2. Mrs Wilkinson is a 45-year-old woman who is attending court after crashing into 3-parked vehicles whilst driving home.

 ❌

Explanation: 1) This case appears to be related to a poor understanding of societal norms. This form of behaviour can be seen in individuals with an intellectual disability. In such cases they tend to show no malice or attempt to avoid discovery or the police. 2) This would appear to represent alcohol dependence or misuse, as it appears the patient has crashed their car unintentionally whilst under the influence. One of the most common complications of alcoholism is road traffic accident.



Question 85

Not answered

Marked out of 4.00

Psychotherapy & Ego Defense Mechanisms

Select the most appropriate defense mechanism used for each of the following cases;

1. Tim Wyatt is a 19-year-old man who is fascinated by competitive, tribal, battle scenarios, where groups of men are pitched against one another and one group emerges victorious. He channels his energy relating to this into his football training, and becomes so successful he is invited to play at a professional level. ❌
2. Mrs Bonfield unexpectedly lost her husband 5 days ago, when he died from a heart attack. They had been married for 28 years. Mrs Bonfield's daughter comes to stay with her, and notices that her mother continues to lay a place at the table for her late husband, and doesn't seem to notice she is doing this. ❌
3. Terry returns from work in a foul mood. He feels his boss has treated him unfairly and he's angry. When the cat gets in his way, he tries to give it a kick, and feels better for having done so. ❌
4. Kitty has recently met Mark, a handsome young man. She feels she is totally in love with him, that everything he does and says is perfect, and that they are totally suited to one another. ❌

Explanation: 1) Sublimation- Potentially destructive or aggressive impulses may be directed towards productive activity that contributes to society 2) Denial is a normal stage of mourning, where making full realisation of a loss is too unbearable 3) Displacement- The anger for the boss is displaced onto the cat in this scenario 4) Idealisation- Falling in love can stimulate idealisation, whereby the more ordinary aspects of the person one is in love with are not noticed.

Question 86

Not answered

Marked out of 4.00

Psychotherapy & Treatment options

Select the most appropriate treatment options for each of the following cases;

1. Amy is a 12-year-old girl who has recently been refusing to go to school. Her father has recently started a new job, which requires him to be abroad quite often, and her mother is feeling emotionally fragile and feels she can't cope with the situation.

 ❌

2. Jessica Stoner is a 19 year old woman who had a disrupted childhood: she was taken into care when she was 1 year old and has lived with several different foster families, making it difficult to sustain each new relationship, disrupting her capacity for consistent attachment. She has been self harming by superficially lacerating her thighs over the past few months.

 ❌

3. Michael is a 28-year-old man who has been feeling low in mood, particularly since the relationship with his girlfriend ended 6 months ago, and he moved to a new city to pursue a new job that requires him to have considerably more responsibilities.

 ❌

4. Stefan presents to his GP with a one-month history of low mood, struggles with low self-esteem, some lack of appetite (although no weight loss). He's not sure what has brought this on and he seems motivated to understand it better and do some collaborative work in talking therapy.

 ❌

Explanation: 1) In this scenario it seems important to work with the whole family, as Amy's school refusal may be a manifestation of difficulties in the whole system 2) MBT (as well as DBT and CAT) has a good evidence base for helping people presenting with features of Emotionally Unstable Personality Disorder. 3) IPT is particularly helpful for role transitions and relationship struggles. 4) Mild-moderate depressive illness can often be helped with Cognitive Behavioural Therapy and is a NICE-recommended treatment.

Question 87

Not answered

Marked out of 5.00

Dementia Management

Choose the single most appropriate treatment for each of the following clinical situations;

1. Harry Whitaker is an 80-year-old man with moderate Alzheimer's Dementia, who over the past few days has been more verbally hostile than usual to staff at his nursing home. ✖
2. Mrs May is 84 years old, has Alzheimer's Dementia, takes Memantine at an adequate dose, and is currently residing in a care home. She is in reasonable physical health. Mrs May has been physically aggressive to staff and visitors for several months, and everyone involved has worked hard to implement behavioural management techniques, with little success. ✖
3. Mr Singh is a 78-year-old man with Parkinson's Dementia whose hostility, agitation and aggression have been causing concern over the past few months. He has been investigated for physical health problems and nothing has been found to suggest he has an acute physical illness. ✖
4. Mr Churchill is a 79 year old man with Alzheimer' dementia. He developed dizziness and bradycardia with donepezil and Galantamine. The family would like him to take antidementia medications. ✖
5. Mrs Dorothy has Alzheimer's disease and she is on Antidementia medication for the last 2 years. She stopped eating as she thought her food was poisoned. She doesn't allow anyone in her family to enter inside her flat, as she believed that they are trying to harm her. ✖

Explanation: 1) Behavioural management should be tried as the first option in dementia patients presenting with challenging behaviour. 2) Memantine is as effective as Risperidone at alleviating difficulties relating to behavioural disturbances, but if Memantine has been trialled and not been effective, it is reasonable to trial Risperidone, taking into account potential adverse effects and associated risk factors. 3) Rivastigmine is associated with moderate improvement in dementia associated with Parkinson's Disease, although there are also unwanted effects (eg nausea, tremor). 4) Memantine can be used in patients who are first tried on acetyl cholinesterase inhibitors and developed side effects on taking them. 5) Risperidone should be used to treat her psychotic symptoms and it is licensed in the UK for the management of behavioural & psychological symptoms of dementia (BPSD).



Question 88

Not answered

Marked out of 4.00

Depression in Older Age

Select the most appropriate diagnosis for each of the following cases;

1. Valerie Watts is a 75-year-old head teacher, who worked beyond the usual retirement age, and finally brought her job to an end 4 months ago. Her family have been in touch with her doctor to say that soon after her retirement she began to become more withdrawn, introspective. She lost her appetite and has lost a stone in weight, and she has told them that she believes her body parts are vanishing, and that she is to blame for 'all that happened', although it's unclear what she is referring to. She appears perplexed and preoccupied.

❌

2. Mr Collon is an 84-year-old retired mechanic. His MMSE score was 27/30. His family say that he seems to have lost his motivation and interest in life: he spends most of his time indoors, staring out of the window, doing very little cooking and not keeping up with friends. Enquiries into his medical history reveal that he had a stroke four months ago, and that he uses a GTN spray for angina.

❌

3. Mrs Tootankamu is 73 years old and has presented to her GP to say she's feeling low in mood; for the past 3 weeks she's been waking up at 4am, feeling hopeless and helpless, she is no longer interested in the world around her and has been thinking about death a lot. She has lost most of her closest friends over the past 5 years, many of them having died from unexpected illness, such as cancer.

❌

4. Mrs Marjorie was referred by her GP for an urgent assessment. She is self-neglecting, isolating herself and has a history of compulsive hoarding. She lives in her home in filthy living conditions.

❌

Explanation: 1) Case 1 has severe depression and psychotic symptoms with mood congruent delusions. 2) Case 2 has depressive features and vascular risk factors like heart disease and recent history of cerebrovascular accident. 3) Case 3 is a Moderate depression includes the two core features (low mood and anhedonia) with four or more related symptoms. 4) Case 4 is Diogenes syndrome. The person presents with self-neglecting behaviour, isolating herself and has a history of compulsive hoarding.



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